



रेल मंत्रालय
MINISTRY OF
RAILWAYS

**NORTHERN RAILWAY ZONE - OFFICIAL INFRASTRUCTURE TENDER
SUBMISSION**

Tender Inviting Authority: Chief Project Manager Electrical Track, Northern Railway
Headquarters, Baroda House, New Delhi - 110001

Tender Notice No.: NR/HQ/W-III/2026/8821-C

Name of Work: Comprehensive Trackside Sub-Station Upgradation and Signaling
Power Automation across the Ghaziabad-Kanpur Railway Section.

SECTION 1: FINANCIAL DECLARATION AND BID SUMMARY

Estimated Value: 493398.34

Bid Amount: 553002.46

Calculated Bid-to-Estimate Ratio: 1.121

Earnest Money Deposit: 11068.00 deposited via RTGS UTR No NRIN202609174412

Performance Bank Guarantee Commitment: 5 percent of Quoted Amount upon Letter
of Acceptance

SECTION 2: VENDOR PROFILE AND CORPORATE HISTORY

Name of Bidding Enterprise: Apex Rail Traction and Power Solutions Pvt. Ltd.

Registered Office: Sector 63, Noida Industrial Area, Uttar Pradesh - 201301

Vendor Age: 1128 Days

Permanent Account Number: BBBB5678L

GSTIN Registration Number: 09BBBBB5678L1Z2

Employee Count: 27

Corporate History and Track Record: Apex Rail Traction and Power Solutions Pvt. Ltd. has been active in regional railway electrical engineering since its incorporation. Our technical team specializes in medium-voltage sub-station automation and trackside feeder installations. We have previously collaborated with divisional railway units on localized signaling modifications. Our workforce comprises 27 certified technicians, site engineers, and administrative personnel capable of executing multi-stage electrification tasks under strict railway safety regulations.

SECTION 3: PLAN OF ACTION AND EXECUTION PHASING

The execution of the Ghaziabad-Kanpur section sub-station upgrade will follow RDSO standards and Indian Railways safety protocols.

Phase 1: Initial Site Reconnaissance and Survey (Weeks 1 to 3): Conducting physical audits of existing sub-station switchboards, transformer bays, and cabling routes. Coordinating with divisional traffic controllers to schedule permitted line blocks and power shutdowns.

Phase 2: Switchgear Replacement and Panel Wiring (Weeks 4 to 10): Dismantling legacy circuit breakers and installing modern vacuum circuit breakers and relay panels. Due to unforeseen site conditions, unstable soil strata, and undocumented legacy wiring configurations encountered during early excavation, a formal Change Order will be required to replace rusted underground earthing strips.

Phase 3: Cable Laying and Interconnection (Weeks 11 to 18): Laying armored power cables through existing masonry trenches. Because aging duct banks frequently exhibit structural blockage, an additional Change Order and contract Amendment will be necessary for manual core cutting and trench rerouting. Furthermore, any variance in cable run lengths discovered during final alignment checks will require an official Revision to material quantities.

Phase 4: System Calibration and Testing (Weeks 19 to 24): Comprehensive relay testing, thermal imaging of busbars, and insulation resistance checks. Additional engineering complexities involving auxiliary power load mismatches will necessitate a subsequent Modification to the control cabinet layout. If integration with secondary signaling loops presents signal attenuation, a further Change Order will be issued. Any mandatory interface adjustments requested by the railway electrical inspector will require an additional contract Amendment. Finally, any adjustment to the final maintenance

handover schedule due to regional weather delays will be governed by a final Change Order document.

SECTION 4: BILL OF MATERIALS AND TECHNICAL SPECIFICATIONS

1. Switchgear Panels: 12 units of 33kV outdoor vacuum circuit breaker panels with numerical protection relays.
2. Power Cables: 2500 meters of XLPE insulated aluminum conductor armored cables.
3. Control Units: 4 sets of programmable logic controller units for automated feeder switching.

SECTION 5: COMMERCIAL JUSTIFICATION AND COMPLIANCE

Our financial bid of 553002.46 accounts for standard material costs, statutory labor wages, and mandatory safety compliances under the Railways Act and EPFO guidelines. All documents are submitted via the e-procurement portal.